

EDUGENIE

Google Gemini Powered Learning Assistant

4. Project Planning

Development Milestones

- Milestone 1 – Model Selection and Architecture.
- Milestone 2 – Core Functionalities Development.
- Milestone 3 – Frontend Development.
- Milestone 4 – Deployment and Functional Testing.

Planned Activities

- Select the AI models and define the application architecture.
- Implement explanation, Q&A, quiz, summary, and learning-path modules.
- Create FastAPI RESTful endpoints and connect them to the modules.
- Build the HTML/CSS user interface.
- Integrate frontend form submission with the backend.
- Run the application locally using Uvicorn.
- Test each major function with representative educational inputs.

Technology Plan

- Programming language: Python.
- Backend: FastAPI.
- Server: Uvicorn.
- Frontend: HTML and CSS.
- Templating: Jinja2.
- Cloud AI: Google Gemini API.
- Local concept explanation model: LaMini-Flan-T5-783M, as specified in the documentation.

Execution Plan

- Prepare the Python environment and dependencies.
- Configure the Gemini API key securely.
- Implement and test backend modules.

- Connect modules to FastAPI endpoints.
- Build and style the frontend.
- Integrate frontend and backend.
- Perform functional testing and prepare project documentation and demonstration.

Source basis: EduGenie project documentation provided by the user.